POLITEKNIK MALAYSIA MERSING • DFP40443 FULL STACK WEB DEVELOPMENT

Assignment Submission & Student Portfolio Management System

A centralized PHP/MySQL web platform enabling seamless digital assignment workflow, real-time AJAX search, and secure portfolio management.

PRESENTERS

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LECTURER

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JTMK Department

CONTEXT & MOTIVATION

Addressing Academic Submission Bottlenecks

Traditional Method Issues

- **Unorganized Submissions:** Missing physical files and untracked emails.
- **Inefficient Verification:** High lecturer workload verifying late submissions.
- **No Portfolio View:** Students lack centralized record of past coursework.

The Digital Solution

- **Central Portal:** Role-based access for students and lecturers.
- **Automated Handling:** Instant file upload, validation, and storage.
- **Live Search & Portfolio:** Fast AJAX lookup and student archive.

BUSINESS & EDUCATIONAL VALUE

Entrepreneurial Innovation & Scalability

Problem Solving

Eliminates manual submission loss and streamlines academic department workflows.

Target Users

Tailored for JTMK students, lecturers, and academic department administrators.

Portal Scalability

Modular design easily expands into a institutional LMS or IT Helpdesk system.

SYSTEM ARCHITECTURE

Full Stack Technology Suite



PHP 8

Backend routing, authentication & file upload logic.



MySQL

Relational database with Foreign Key relationships.



Bootstrap 5

Responsive user interface and clean form components.



AJAX & JS

Asynchronous fetch API for live dynamic searching.

RELATIONAL DATABASE SCHEMA

`assignment_system` Database ERD

Table: `users`

id (PK, Auto-Increment)

full_name (VARCHAR)

email (VARCHAR, Unique)

password (VARCHAR, Hashed)

role (ENUM: admin/student)

Table: `categories`

id (PK, Auto-Increment)

category_name (VARCHAR)

description (TEXT)

created_at (TIMESTAMP)

Table: `assignments`

id (PK, Auto-Increment)

title (VARCHAR)

category_id (FK → categories.id)

created_by (FK → users.id)

Table: `submissions`

id (PK, Auto-Increment)

user_id (FK → users.id)

assignment_id (FK → assignments.id)

file_path (VARCHAR)

SYSTEM SECURITY

Authentication & Data Protection

Password Encryption

Uses PHP `password_hash()` BCRYPT algorithm for secure user authentication.

Role-Based Access (RBAC)

Enforces PHP session checks to separate Student routes from Admin control routes.

Prepared Statements

All database transactions use parameterized queries to stop SQL injection attacks.

STUDENT MODULE

Dynamic AJAX Search & Upload Workflow

STEP 01

AJAX Live Search

Instant assignment filter via `search_assignments.php` without page reloads.

STEP 02

File Upload Validation

Enforces size restrictions (≤5MB) and format checks (PDF, DOCX, ZIP).

STEP 03

Student Portfolio View

Centralized archive allowing students to verify submission records anytime.

LECTURER / ADMIN MODULE

Administrative Management Tools

Category Management

Create and manage subject categories for streamlined coursework grouping.

Assignment Publishing

Publish assignments with deadlines and instructions directly to students.

Submission Downloads

Inspect submission timestamps and download attached files for evaluation.

TEAM COLLABORATION

Task Allocation & Version Control

NS **Nurul Safiyyah Aliyah**
Backend Lead & Admin Module

- Developed Admin CRUD endpoints
- SQL JOIN queries & ERD setup
- System testing & database exports

AB **Alya Batrisyia**
Frontend & Student Module Lead

- AJAX Live Search implementation
- File upload validation & UI styling
- Main GitHub repo maintainer

GitHub Repository: aliyah1126/mini_project_2

Shared Commit History & Branch Management

RUBRIC COMPLIANCE VERIFIED • CLO3 (A2, PLO10)

Thank You!

Questions & Demonstration

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